Test 11 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).
A four-function calculator may be used **only** on items marked *calculator*.
The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.
Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Match each written description with the expression it represents.

	(x + 6)/2	6x + 6	x - 6	x/6
the quotient of a number x and 6	■	■	■	■
a number x decreased by 6	■	■	■	■
the sum of a number x and 6, divided by 2	■	■	■	■

Show your work

matching table

2. A data set is shown.

7, 9, 10, 11, 13

The value 37 is added to the data set. What is the range of the new data set?

A. 33
B. 37
C. 6
D. 30

Show your work

multiple choice

3. Which fraction is equivalent to 45%, written in simplest form?

A. 9/2
B. 9/20
C. 20/9
D. 9/10

Show your work

calculator · multiple choice

4. Each expression is evaluated for x = 3. Match each expression with its value.

	0	18	21	54
6x + 3	■	■	■	■
6x^2	■	■	■	■
3 - x	■	■	■	■

Show your work

matching table

5. Select **all** the expressions equal to 3/8.

- ■ 4/6
- ■ 3/4 + 1/2
- ■ 3/4 ÷ 1/2
- ■ 3/8
- ■ 1/2 × 3/4
- ■ 3/4 × 1/2

Select all that apply.

Show your work

calculator · multiselect

6. A ticket costs $7.12 and 8 tickets are bought.

Select **all** the expressions that give the total cost in dollars.

- ■ 8 × 7.12
- ■ 56.96 ÷ 8
- ■ 7.12 + 8
- ■ 56.96 ÷ 1
- ■ 56.96 × 8
- ■ 7.12 × 8

Select all that apply.

Show your work

calculator · multiselect

7. 20 students is 50% of the students in a grade.

How many students are in the grade?

A. 70
B. 2000
C. 1000
D. 40

Show your work

multiple choice

8. A change of more than 8 degrees from 0 sets off an alarm.

Select **all** the readings that set off the alarm.

- ■ -10
- ■ 9
- ■ -8
- ■ 4
- ■ 1
- ■ -5

Select all that apply.

Show your work

calculator · multiselect

9. Use the drop-down menus to give each solution.

If x + 16 = 31, then x = [[1]].
If x - 18 = 15, then x = [[2]].

Blank (1) choices: 15 | 16 | 31 | 47
Blank (2) choices: -3 | 15 | 18 | 33

Show your work

editing task choice

10. Select **all** the numbers that are common factors of 24 and 36.

- ■ 2
- ■ 9
- ■ 4
- ■ 3
- ■ 5
- ■ 6

Select all that apply.

Show your work

calculator · multiselect

11. Store A sells 6 bottles of water for $3.00. Store B sells 12 bottles for $8.00.

What is the unit price at Store A, in dollars per bottle?

A. 9
B. 2
C. 0.6667
D. 0.5

Show your work

multiple choice

12. Select **all** the questions below that would generate numerical data with variability.

Circle every correct choice:

- ■ How many minutes did each sixth grader spend on the bus this morning?
- ■ How many books did each student in the class read last month?
- ■ How tall is the tallest student in my class?
- ■ What time does the first bell ring at my school?
- ■ How old is my teacher?

Show your work

selectable hot text

13. Use the drop-down menus to complete the solutions.

If x + 2.54 = 5.36, then x = [[1]].
If x - 2.54 = 2.82, then x = [[2]].

Blank (1) choices: 2.54 | 2.82 | 7.9
Blank (2) choices: 0.28 | 2.82 | 5.36

Show your work

editing task choice

14. Gabriel's prepaid transit card shows a balance of 0 dollars.

What does the 0 mean in this situation?

A. The card has been used 15 times.
B. The card is worth more than it was yesterday.
C. The card has $15 of credit remaining.
D. There is no money left on the card and no amount is owed.

Show your work

calculator · multiple choice

15. What is the value of -81 ÷ (9)?

A. 9
B. -90
C. -7
D. -9

Show your work

calculator · multiple choice

16. Use the drop-down menus to give each solution.

If 4x = 44, then x = [[1]].
If x ÷ 5 = 11, then x = [[2]].

Blank (1) choices: 4 | 11 | 44 | 176
Blank (2) choices: 2 | 5 | 11 | 55

Show your work

editing task choice

17. A data set is shown.

6, 10, 12, 17, 17, 22

Match each measure with its value.

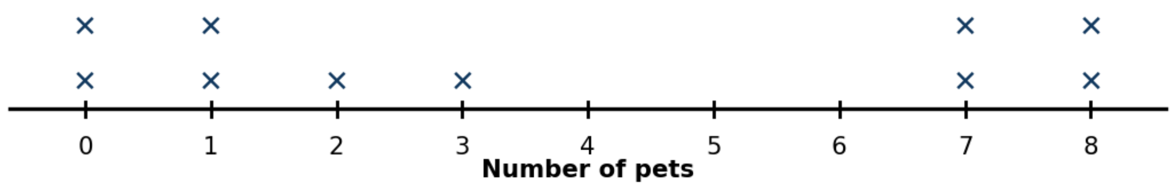
	14	14 1/2	16	18
Mean	■	■	■	■
Median	■	■	■	■
Range	■	■	■	■

Show your work

matching table

18. The line plot shows number of pets.

How many students are represented on the line plot? Enter your answer.



Answer: ______________________________

Show your work

equation editor

19. Match each rate with its unit rate (the amount for exactly 1 unit).

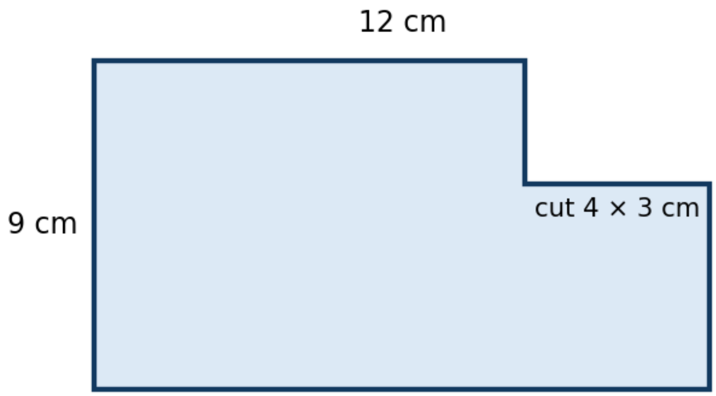
	4	8	12	14
16 miles in 4 hours	■	■	■	■
16 dollars for 2 pounds	■	■	■	■
72 pages in 6 minutes	■	■	■	■

Show your work

matching table

20. Use the drop-down menus to find the area of the figure, in square centimeters.

The area of the whole rectangle is [[1]] square centimeters, and the area of the cut-out corner is [[2]] square centimeters.



Blank (1) choices: 21 | 42 | 96 | 108
Blank (2) choices: 7 | 12 | 14 | 36

Show your work

editing task choice

21. A parking lot has 6 cars and 4 trucks.

Use the drop-down menus to write the ratio of cars to trucks in simplest form.

[[1]] : [[2]]

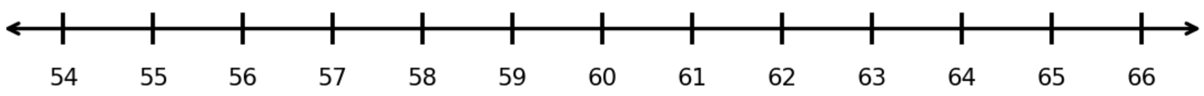
Blank (1) choices: 2 | 3 | 4 | 6
Blank (2) choices: 2 | 3 | 4

Show your work

editing task choice

22. A carry-on bag must weigh no more than 60 pounds.

Plot the boundary value of the solution on the number line.



Mark your answer on the diagram above.

Show your work

graphing (grid)

23. For each value of x, decide whether x ≥ 10 is true or false.

	True	False
8	■	■
10	■	■
11	■	■
19	■	■

Show your work

matching table

24. Match each point with the quadrant it is located in.

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
A (3, 4)	■	■	■	■
D (5, -2)	■	■	■	■
C (-4, -3)	■	■	■	■

Show your work

matching table

25. Match each triangle with its area, in square feet.

	14	20	27	30
base 5 feet, height 8 feet	■	■	■	■
base 9 feet, height 6 feet	■	■	■	■
base 7 feet, height 4 feet	■	■	■	■

Show your work

matching table

26. A number has the prime factorization 3^2 × 5.

What is the number?

A. 90
B. 45
C. 22
D. 11

Show your work

calculator · multiple choice

27. A box plot of a data set has the five-number summary shown below the table.

Match each part of the summary with its value.

	2	4	9	14
Minimum	■	■	■	■
Median	■	■	■	■
Upper quartile	■	■	■	■

Show your work

matching table

28. Select **all** the expressions with a value of 2.4.

- 2.5 × 1
- 2.4 × 10
- 24 ÷ 10
- 2.4 ÷ 0.1
- 0.4 × 6
- 1.2 × 2

Select all that apply.

Show your work

calculator · multiselect

29. Select **all** the numbers with an absolute value greater than 4 1/2.

- -1
- -4 1/2
- 3
- 7
- -6 1/2
- -2 1/2

Select all that apply.

Show your work

calculator · multiselect

30. Select **all** the expressions equal to 132 + 84.

- 13(11 + 7)
- 12(11 + 7)
- 11(12 + 7)
- 12 × 18
- 12(11 + 8)
- 132 + 84

Select all that apply.

Show your work

calculator · multiselect

31. Select **all** the expressions with a value of 36.

- ■ 6^2
- ■ 6^3
- ■ 36 × 1
- ■ 6 × 6
- ■ 6 × 2
- ■ 2^6

Select all that apply.

Show your work

calculator · multiselect

32. Select **all** the expressions with a value of 132.

- ■ 132 × 1
- ■ 2 × 2 × 3 × 11
- ■ 132 × 2
- ■ 2^2 × 3 × 11
- ■ 2^2 × 3^2 × 11
- ■ 132 + 1

Select all that apply.

Show your work

calculator · multiselect

33. Match each expression with an equivalent expression.

	13x	21x	7x	7x + 21
7(x + 3)	■	■	■	■
7x + 6x	■	■	■	■
7x + 3 - 3	■	■	■	■

Show your work

matching table

34. What is the value of 14 - (-3)?

A. -17
B. 11
C. 20
D. 17

Show your work

calculator · multiple choice

35. Select **all** the numbers that are less than -1 1/2.

- 0
- 2
- -1 3/4
- -2 1/2
- 3/4
- -1/2

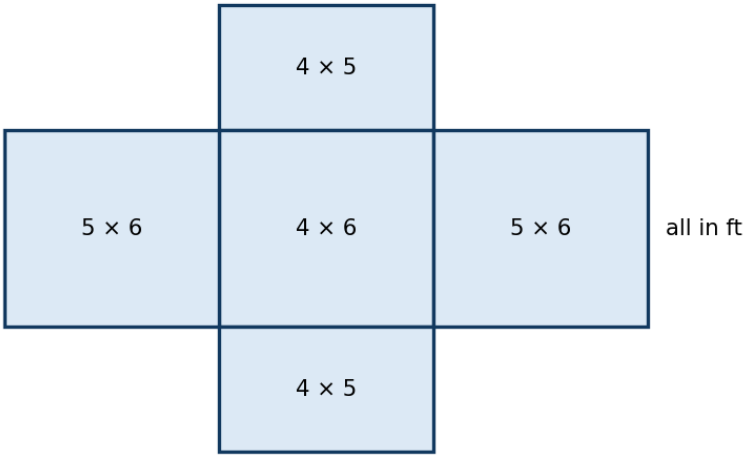
Select all that apply.

Show your work

calculator · multiselect

36. Use the drop-down menus to find the surface area of the prism from its net.

The two 4 by 5 faces have a total area of [[1]] square feet, and the two 4 by 6 faces have a total area of [[2]] square feet.



Blank (1) choices: 9 | 20 | 40 | 80
Blank (2) choices: 10 | 24 | 48 | 96

Show your work

editing task choice

37. A rectangular garden is drawn on a coordinate plane with vertices (1, -4), (9, -4), (9, 2) and (1, 2). Each unit is 1 meter.

What is the perimeter of the garden, in meters? Enter your answer.

Answer: ______________________________

Show your work

equation editor

38. Match each right rectangular prism with its volume, in cubic inches.

	36	48	60	66
2 by 5 by 6 inches	■	■	■	■
2 by 2 by 9 inches	■	■	■	■
4 by 4 by 3 inches	■	■	■	■

Show your work

matching table

Grade 6 Mathematics Reference Sheet

Customary Conversions	
1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions	
1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions	
1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula	
Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume